

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 3

Article Review

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO 4: Students will be able to analyze and evaluate advanced deep learning techniques and architectures, interpret research findings, and apply appropriate models to real-world problems. They will be able to critically review research articles, compare methodologies, and justify suitable deep learning solutions. (BL-6)

Assessment Tool Weightage: 30%

QUESTIONS:

Q1. Article Identification and Summary

Select a recent peer-reviewed research article related to Deep Learning, such as CNNs, Transfer Learning, ResNet, DenseNet, RNNs, LSTM, Autoencoders, or other advanced architectures. Provide the article title, authors, publication venue, year, and research objective. Summarize the main problem addressed, proposed approach, and key findings in your own words.

Q2. Problem Statement and Motivation

Explain the real-world problem addressed by the selected article and analyze why the problem requires a deep learning solution. Identify the limitations of existing approaches discussed by the authors and explain the motivation for the proposed method.

Q3. Methodology and Architecture Review

Critically review the methodology used in the article. Describe the dataset, preprocessing, model architecture, important layers or components, training strategy, and evaluation metrics. Illustrate the proposed architecture or workflow and explain the role of its major components.

Q4. Results, Comparison and Critical Analysis

Analyze the experimental results reported in the article. Compare the proposed method with the baseline or existing methods using the reported performance measures. Discuss the strengths, limitations, computational considerations, and validity of the results. Identify at least two possible improvements or future research directions.

Q5. Application and Personal Evaluation

Explain how the reviewed deep learning method can be applied to a practical industry problem. Propose a suitable implementation or extension and justify your choices. Conclude the review with your overall evaluation of the article, including its contribution, practical relevance, and scope for future work.

Article Review Submission Guidelines

- Choose one recent and relevant peer-reviewed research article in Deep Learning.
- Attach or provide the complete citation/reference of the selected article.
- The review must be written in the student's own words and should demonstrate critical analysis.
- Include architecture diagrams, tables, graphs, or screenshots where they improve explanation.
- Use appropriate academic referencing and avoid plagiarism.
- Recommended review length: 4–6 pages, excluding the reference page.

Code of Conduct:

I Haresh Kumar N L (192425009) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Article Understanding & Summary	20%	Comprehensive and accurate understanding of the article, problem, objectives, and contributions.	Good understanding with minor omissions or inaccuracies.	Basic understanding; important details are missing.	Poor understanding or significant misinterpretation .
Critical Review of Methodology	25%	Thoroughly evaluates methodology, architecture, datasets, training, and evaluation choices.	Appropriate analysis with minor gaps.	Limited analysis with several unexplained aspects.	Little or no meaningful analysis of methodology.
Analysis of Results	20%	Accurately interprets results, comparisons, metrics, strengths, and limitations.	Good interpretation with reasonable comparison.	Basic interpretation with limited comparison.	Incorrect, unsupported, or missing interpretation.

Application & Proposed Improvements	20%	Practical, innovative application with well-justified improvements and future directions.	Relevant application and feasible improvements .	Basic application with limited justification.	Unclear, impractical, or unsupported proposal.
Presentation & Referencing	15%	Well-organized, professional presentation with clear diagrams and correct academic referencing.	Clear presentation with minor formatting or referencing issues.	Adequate presentation with inconsistent formatting/referencing.	Poor organization, unclear presentation, or inadequate referencing.

Suggested Article Review Structure

1. Article Citation
2. Abstract / Executive Summary
3. Research Problem and Motivation
4. Methodology and Deep Learning Architecture
5. Dataset and Experimental Setup
6. Results and Comparative Analysis
7. Strengths and Limitations
8. Proposed Improvements / Future Scope
9. Practical Industry Application
10. Conclusion and References

Answers:

ARTICLE REVIEW

1. Article Citation

Title of the Research Article:

Methodology for Evaluating the Generalization of ResNet

Authors:

Anan Du, Qing Zhou, and Yuqi Dai

Publication Venue:

Applied Sciences

Year of Publication:

2024

Volume and Issue:

Volume 14, Issue 9

Article Number:

3951

DOI:

10.3390/app14093951

Research Area:

Deep Learning, Convolutional Neural Networks (CNNs), ResNet, Model Generalization, Computer Vision

Complete Citation

Du, A., Zhou, Q., & Dai, Y. (2024). *Methodology for Evaluating the Generalization of ResNet*. Applied Sciences, 14(9), 3951. <https://doi.org/10.3390/app14093951>

The selected article is a recent peer-reviewed research article published in 2024. It focuses on an important issue in deep learning: determining whether a trained CNN model can generalize effectively to previously unseen data. The authors use ResNet as the main architecture for investigating this problem.

2. Abstract / Executive Summary

Deep learning models have become highly successful in computer vision because of their ability to automatically learn useful features from large amounts of image data. Convolutional Neural Networks (CNNs), in particular, have achieved strong performance in image classification and many other visual tasks. However, high performance on training data does not always guarantee that a model will perform equally well on unseen data. This ability to perform effectively on new data is known as **generalization**.

The reviewed article investigates the problem of evaluating the generalization ability of CNN models. The researchers focus on ResNet architectures because ResNet is an important deep learning architecture widely used in practical computer vision applications. The authors identify a limitation in previous generalization-evaluation approaches: many existing studies were tested using small or simplified sets of CNN models. This can make it difficult to determine whether an evaluation method will remain reliable for larger and more realistic deep learning architectures.

To address this issue, the researchers propose an evaluation method based on **Intersection over Union (IoU)**. The method measures the overlap between the features learned by the CNN and the actual location of the relevant object in an image. **Class Activation Maps (CAMs)** are used to visualize and identify the areas of an image that contribute to the model's prediction.

For experimental validation, the authors constructed a large collection of ResNet models. The model set contained **2,000 CNN models** trained using a subset of ImageNet. Different commonly used hyperparameters were systematically changed to create models with different generalization abilities. The experiments used ResNet18, ResNet34, ResNet50, and ResNet101 architectures.

The results showed a strong relationship between the proposed IoU-based measure and the actual generalization performance of the models. The reported Pearson correlation coefficient was greater than **0.8**, indicating a strong correlation. The researchers also performed additional experiments to examine the robustness of the proposed approach. Overall, the article demonstrates that analyzing whether a CNN focuses on task-relevant image regions can provide useful information about its ability to generalize.

3. Research Problem and Motivation

3.1 Research Problem

One of the major challenges in deep learning is ensuring that a model performs well on data that it has never seen during training. A model can achieve very high training accuracy while still performing poorly on unseen data. This problem is closely related to **overfitting**.

In practical applications such as autonomous driving, medical image analysis, industrial inspection, and other computer vision systems, generalization is extremely important. A model deployed in the real world will encounter images that are different from the exact images used during training. Therefore, simply observing training accuracy is not sufficient to determine whether the model is reliable.

The authors explain that evaluating the generalization ability of deep CNNs is difficult because modern networks are highly parameterized. The number of model parameters can be much larger than the number of training samples. Consequently, understanding why one model generalizes better than another remains a challenging research problem.

3.2 Limitations of Existing Approaches

The article discusses two broad categories of approaches for evaluating model generalization.

The first category is based on **theoretical analysis**. These approaches use concepts from statistical learning theory to estimate or bound generalization error. However, theoretical approaches can depend on strong assumptions and may not accurately describe the behavior of modern deep neural networks in practical situations.

The second category is based on **empirical analysis**. These methods attempt to predict generalization using characteristics of model weights, intermediate representations, or other learned features. Although these methods can be useful, previous studies were often evaluated on small or simplified collections of models.

The authors specifically point out that previous model sets were often based on datasets such as CIFAR and SVHN, where images have relatively low resolution and the models can be smaller than those commonly used in practical applications. This creates a gap between experimental research and real-world deep learning systems.

3.3 Motivation for the Proposed Method

The main motivation of the research is therefore to create a more practical method for evaluating CNN generalization. Instead of examining only model parameters, the authors investigate **what visual features the model is actually learning**.

The central idea is that a model should ideally focus on the relevant object or region when making a classification decision. If the model learns features that correspond closely to the actual object location, this may indicate that the model has learned meaningful and task-relevant information.

Based on this idea, the researchers use Class Activation Maps to identify the spatial distribution of learned features and IoU to measure their overlap with the actual object region. This provides a visual and quantitative approach to studying model generalization.

4. Methodology and Deep Learning Architecture

4.1 Overall Methodology

The methodology consists of several major stages:

Image Dataset → ResNet Training → Class Activation Map Generation → Feature Region Extraction → IoU Calculation → Generalization Evaluation

The researchers first train different ResNet models using an ImageNet subset. After training, the models are analyzed using Class Activation Maps. CAMs show which areas of an input image contribute to the model's classification decision.

The learned feature region is then compared with the actual location of the relevant object. IoU is used to quantify the amount of overlap between these two regions.

A higher IoU indicates that the learned feature distribution is more closely aligned with the actual object location. According to the proposed methodology, this provides an indication of stronger generalization capability.

4.2 ResNet Architecture

ResNet, or Residual Network, is a deep CNN architecture designed to make the training of very deep networks easier. Its important characteristic is the use of **residual connections**.

Instead of requiring every layer to directly learn a complete transformation, residual learning allows the network to learn a residual mapping. Shortcut connections help information and gradients flow through deeper networks.

The research uses four ResNet architectures:

- **ResNet18**
- **ResNet34**
- **ResNet50**
- **ResNet101**

These architectures provide different depths and therefore allow the researchers to evaluate the proposed generalization method across networks with different levels of complexity.

4.3 Class Activation Maps

A major component of the proposed methodology is the **Class Activation Map (CAM)**.

A CAM represents the importance of different spatial regions of an image for a particular classification decision. It provides a visual representation of where the CNN is focusing when it predicts a particular class.

The authors also consider Grad-CAM-based approaches because they can generate activation maps without requiring the same architectural modifications associated with the original CAM approach.

This makes activation maps useful for examining whether the model has learned features that correspond to the actual object rather than unrelated background information.

4.4 IoU-Based Evaluation

The central contribution of the paper is the use of **Intersection over Union (IoU)** to evaluate the ratio of task-relevant features.

The process can be represented as:

Actual Object Location + Learned Feature Distribution → Spatial Overlap → IoU Value

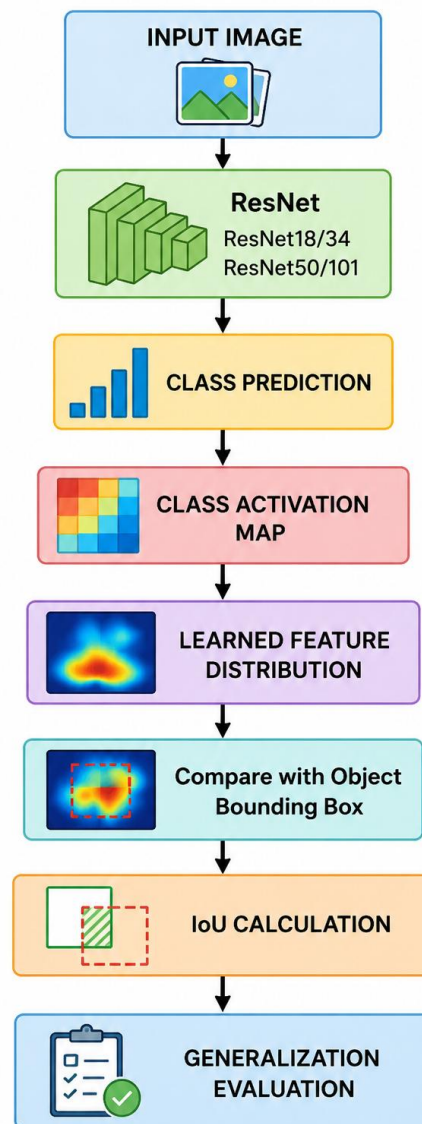
IoU is calculated using the intersection between the learned feature region and the actual object region divided by their union.

The value ranges from **0** to **1**:

- **IoU close to 0**: little overlap between learned features and the actual object.
- **IoU close to 1**: strong overlap between learned features and the actual object.

The authors therefore use the IoU value as an indicator of how closely the model's learned features correspond to task-relevant regions.

Proposed Architecture / Workflow



The workflow demonstrates that the proposed method does not simply evaluate classification accuracy. It also examines whether the model is learning spatially meaningful information.

5. Dataset and Experimental Setup

5.1 Dataset

The researchers used a **subset of ImageNet** for their experiments. ImageNet is a large-scale image dataset commonly used for evaluating computer vision models.

The use of an ImageNet subset makes the experimental setup more representative of practical computer vision applications than experiments based only on very small, low-resolution datasets.

5.2 Model Set

One of the major strengths of the study is the size and diversity of its model set.

The researchers constructed approximately **2,000 CNN models** by systematically changing commonly used hyperparameters. The architectures included:

Model	Description
ResNet18	Shallower ResNet architecture
ResNet34	Medium-depth ResNet
ResNet50	Deeper ResNet architecture
ResNet101	Very deep ResNet architecture

The authors also varied important training factors, including the optimization algorithm, regularization coefficient, and model structure. SGD and Adam were used as optimization algorithms, while different regularization settings were considered.

5.3 Training Strategy

The models were trained using the ImageNet subset. The researchers established an upper training limit of **150 epochs**. Their observations indicated that approximately 90% of the models reached the termination criteria within the first 50 epochs.

The large model collection allowed the researchers to obtain models with different levels of generalization performance. This was important because an evaluation method cannot be properly tested if all models have almost identical behavior.

5.4 Evaluation Metric

The main statistical evaluation metric was the **Pearson correlation coefficient**.

The researchers compared the IoU-based measure with the actual **generalization gap**.

The generalization gap can be understood as the difference between training performance and test performance. A smaller gap generally indicates better generalization.

The purpose of Pearson correlation analysis was to determine whether changes in the proposed IoU measure corresponded to changes in the actual generalization performance of the models.

6. Results and Comparative Analysis

6.1 Main Results

The main experimental finding was that the proposed IoU-based evaluation method showed a **strong correlation with model generalization performance**.

The reported Pearson correlation coefficient was **greater than 0.8**, demonstrating a strong relationship between the proposed feature-overlap measurement and actual model generalization.

This is important because it indicates that the spatial relationship between learned CNN features and the actual object can provide useful information about how well a model is likely to generalize.

6.2 Comparison with Existing Evaluation Approaches

The researchers compared the IoU-based approach with other generalization assessment methods.

The study reports that the IoU-based approach produced strong results and was computationally efficient because it can evaluate a trained model using the training data and the trained model itself.

The experiments also showed that the relationship between the evaluation measures and generalization became stronger when models with higher training accuracy were considered.

The results across different ResNet architectures demonstrated that the proposed approach was not limited to one particular network depth. The method showed useful correlations for ResNet18, ResNet34, ResNet50, and ResNet101, although some variations were observed between architectures.

Comparative Analysis

Aspect	Proposed IoU Method	Traditional/Other Approaches
Main idea	Measures overlap of learned features and object location	Uses model characteristics or theoretical measures
Interpretability	High because feature regions can be visualized	Often more difficult to interpret
Model requirement	Uses trained model and training data	May require additional calculations or assumptions
Computational consideration	Relatively efficient according to the study	Some approaches can require significant computation

Aspect	Proposed IoU Method	Traditional/Other Approaches
Validation	Tested on 2,000 ResNet models	Previous work often used smaller/simplified model sets
Main limitation	Requires bounding-box annotations	Limitations vary by method

6.3 Robustness

The researchers also investigated whether the proposed approach remained effective when different CAM methods were used.

They tested alternative approaches including **GradCAM++** and **SmoothGradCAM++**. The reported experiments supported the robustness of the proposed IoU-based evaluation approach against variations in the CAM algorithm.

6.4 Critical Analysis of Results

The results are convincing because the researchers did not rely on only one model. Instead, they created a large model set containing approximately 2,000 ResNet models.

Another positive aspect is the use of multiple ResNet depths. Testing ResNet18 through ResNet101 makes the evaluation more meaningful because it demonstrates that the approach is not designed only for a single architecture.

However, the results should not be interpreted as proving that IoU can completely replace all other methods of evaluating generalization. The experiment is focused primarily on image classification and uses a specific dataset and model family. Therefore, additional validation would be required for other architectures, datasets, and computer vision tasks.

7. Strengths and Limitations

7.1 Strengths

1. Large-Scale Model Evaluation

A major strength is the construction of a model set containing approximately 2,000 CNN models. This provides a stronger experimental basis than evaluating the method on only a few networks.

2. Use of Practical ResNet Architectures

The research evaluates ResNet18, ResNet34, ResNet50, and ResNet101. These architectures represent different network depths and make the experiments more relevant to real-world deep learning.

3. Explainable Evaluation

The use of Class Activation Maps provides a visual explanation of the features used by the model. This is useful because deep learning models are often treated as black boxes.

4. Strong Experimental Correlation

The reported Pearson correlation coefficient above 0.8 demonstrates a strong relationship between the IoU measure and model generalization performance.

5. Computational Efficiency

The authors state that the IoU calculation requires the training data and a trained model, making the approach computationally efficient compared with some more resource-intensive approaches.

7.2 Limitations

1. Dependence on Bounding Boxes

The most important limitation identified by the authors is that the proposed IoU calculation requires bounding-box annotations. Creating these annotations manually can be time-consuming and expensive.

2. Limited Task Scope

The research primarily investigates image classification using ResNet models. Therefore, the results may not automatically apply to every deep learning task.

3. Dataset Limitation

The model set was constructed using a subset of ImageNet. Although this is more realistic than some smaller datasets, testing on additional datasets would provide stronger evidence.

4. Limited Number of Categories

The study uses a dataset containing 20 categories. The authors acknowledge that this differs from large-scale classification settings with approximately 1,000 categories and may influence some of the observed results.

8. Proposed Improvements / Future Scope

Based on the limitations and findings of the research, several improvements can be proposed.

8.1 Automatic Bounding-Box Generation

The most direct improvement is to reduce dependence on manually created bounding boxes. An automatic object-detection or segmentation model could be used to generate approximate object regions.

This would reduce annotation effort and make the proposed evaluation method easier to apply to large datasets.

8.2 Evaluation Across More Datasets

Future research should evaluate the method using different datasets from different application areas. For example, medical images, industrial inspection images, traffic images, and satellite images could be considered.

Testing on multiple datasets would determine whether the relationship between IoU and generalization remains consistent across different visual domains.

8.3 Evaluation of More Deep Learning Architectures

The current research focuses on ResNet models. Future work could investigate whether the same methodology works with architectures such as DenseNet, EfficientNet, Vision Transformers, and other modern computer vision networks.

This would help determine whether the proposed approach is a general deep learning evaluation framework rather than a ResNet-specific technique.

8.4 Automated Model Selection

Another potential application is automated model selection. If IoU can reliably indicate generalization, it could be integrated into a model-development pipeline to identify models that are likely to perform better on unseen data.

Instead of selecting a model only based on training accuracy, developers could consider both accuracy and task-relevant feature alignment.

9. Practical Industry Application

The proposed method can be applied to several real-world industries where reliable computer vision systems are required.

9.1 Industrial Quality Inspection

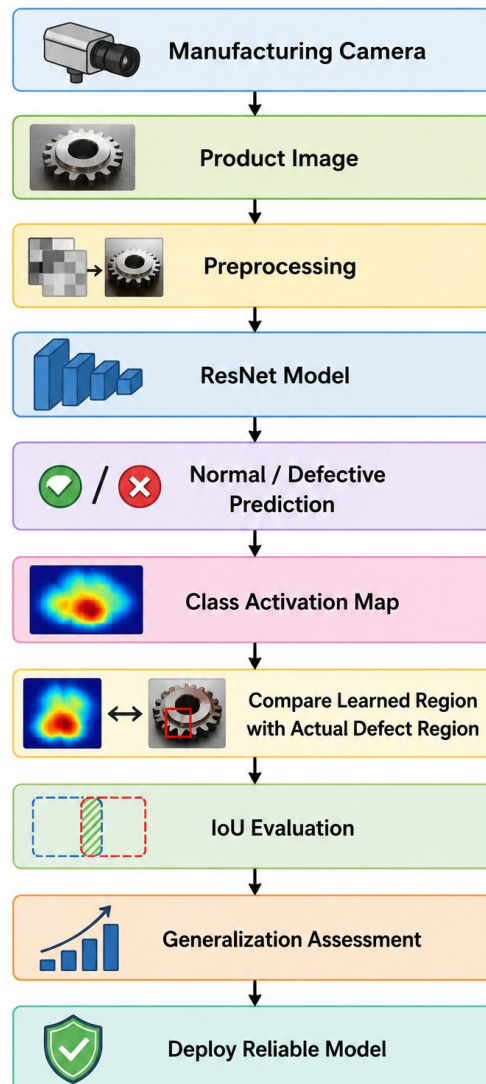
One practical application is **automated industrial defect inspection**.

In manufacturing, cameras can capture images of products such as metal components, electronic boards, automobile parts, or manufactured surfaces. A CNN can then classify the product as normal or defective.

However, high training accuracy does not necessarily mean that the model will work correctly on products produced under different lighting conditions, camera positions, or manufacturing batches.

The proposed methodology could be used to examine whether the CNN is focusing on the actual product or defect region. Class Activation Maps could identify the regions used by the model, while IoU could measure how closely those regions correspond to the relevant product or defect area.

Proposed Industry Workflow



This approach could help engineers identify models that are achieving high accuracy for the wrong reasons. For example, if a model focuses heavily on the background instead of the product surface, its performance may decrease when the production environment changes.

9.2 Autonomous Driving

Another possible application is autonomous driving. A vehicle vision system must recognize cars, pedestrians, road signs, and other objects under changing environmental conditions.

A generalization evaluation method could help determine whether the model is learning meaningful object features rather than relying on irrelevant environmental characteristics.

9.3 Medical Image Analysis

The methodology could also be investigated in medical imaging. A deep learning model may be trained to identify abnormalities in X-rays, CT images, or other medical images.

In this context, it is especially important to determine whether the model is focusing on the clinically relevant region. An IoU-based feature analysis could potentially help researchers evaluate whether the model's learned features correspond to the target region.

10. Conclusion and References

10.1 Conclusion

The reviewed article, *Methodology for Evaluating the Generalization of ResNet*, addresses an important challenge in deep learning: determining whether a CNN model can generalize effectively beyond its training data.

The authors propose a new perspective based on the relationship between the features learned by a CNN and the actual location of relevant objects in an image. Class Activation Maps are used to visualize the model's learned feature distribution, while Intersection over Union is used to quantify the overlap between the learned features and the actual object region.

The experimental study is particularly valuable because it uses approximately 2,000 ResNet models with different architectures and hyperparameter configurations. ResNet18, ResNet34, ResNet50, and ResNet101 are included, allowing the proposed methodology to be examined across networks with different depths.

The results indicate a strong correlation between the proposed IoU measure and actual model generalization, with a reported Pearson correlation coefficient greater than 0.8. The researchers also demonstrate that the method remains robust when different CAM approaches are considered.

From a critical perspective, the research has several strengths, including its large-scale model evaluation, use of practical ResNet architectures, visual interpretability, and computational efficiency. However, the dependence on bounding-box annotations and the focus on a particular dataset and image-classification setting limit the immediate generalization of the method to all deep learning applications.

Overall, the article makes a useful contribution to deep learning research by showing that the spatial relationship between learned features and task-relevant regions can provide information about model generalization. The approach could have practical value in areas such as industrial inspection, autonomous driving, and medical image analysis. Future improvements such as automatic region annotation, evaluation across more datasets, and testing with newer deep learning architectures could make the methodology more broadly applicable.

10.2 Reference

Du, A., Zhou, Q., & Dai, Y. (2024). *Methodology for Evaluating the Generalization of ResNet*. **Applied Sciences**, 14(9), 3951. <https://doi.org/10.3390/app14093951>